

Companies App API Documentation - Complete Reference

Base URL

/api/companies/company/buses/

Authentication Notes

- **All endpoints require authentication** (JWT token)
 - **All endpoints require Company Owner role**
 - Regular users cannot access any bus management endpoints
 - Company owners can only access buses belonging to their own company
-

Endpoint 1: List Buses

GET /api/companies/company/buses/

Allowed Access: Company Owner only (authenticated + owner role)

Query Parameters (pagination):

- `page` (integer): Page number (default: 1)
- `page_size` (integer): Items per page (default: 20)

Response (200 OK):

json

```
{
  "count": 15,
  "next": "http://api.example.com/api/companies/company/buses/?page=2",
  "previous": null,
  "results": [
    {
      "id": 5,
      "company": 3,
      "name": "Bus 101 - VIP Express",
```

```
    "type": "VIP",
    "seat_count": 50,
    "created_at": "2026-01-15T10:30:00Z"
  },
  {
    "id": 6,
    "company": 3,
    "name": "Bus 102 - Standard",
    "type": "Standard",
    "seat_count": 44,
    "created_at": "2026-01-20T14:15:00Z"
  }
]
```

Response Fields:

- id: Unique bus identifier
- company: ID of the company that owns this bus (read-only, auto-assigned)
- name: Bus name or identifier (e.g., "Bus 101 - VIP Express")
- type: Bus type (e.g., "VIP", "Standard", "Economy")
- seat_count: Total number of seats on the bus (minimum 10)
- created_at: ISO 8601 timestamp of when bus was added

Special Behavior:

- Only returns buses belonging to the authenticated owner's company
- Automatic filtering via `get_queryset()`

Endpoint 2: Create Bus

POST /api/companies/company/buses/

Allowed Access: Company Owner only (authenticated + owner role)

Request Body:

json

```
{  
  "name": "Bus 103 - Luxury",  
  "type": "Luxury",  
  "seat_count": 48  
}
```

Field Descriptions:

- name (string, required): Bus name or identifier (max length as per model)
- type (string, required): Type/category of the bus (e.g., VIP, Standard, Economy, Luxury)
- seat_count (integer, required): Number of seats (minimum validation may apply)

Validation Rules:

- Seat count must be positive integer
- Seat count should be at least 10 (for trip creation compatibility)
- Name must be unique within the company (if model has unique_together constraint)

Automatic Fields:

- company: Automatically set to the authenticated owner's company
- created_at: Automatically set on creation

Response (201 Created):

json

```
{  
  "id": 7,  
  "company": 3,
```

```
"name": "Bus 103 - Luxury",
"type": "Luxury",
"seat_count": 48,
"created_at": "2026-05-30T10:00:00Z"
}
```

Error Responses:

- **400 Bad Request:**
 - Missing required fields
 - Invalid seat_count (negative, zero, or less than minimum)
 - Duplicate bus name (if unique constraint exists)
 - **401 Unauthorized:** Missing or invalid JWT token
 - **403 Forbidden:** User is not a company owner
-

Endpoint 3: Retrieve Single Bus

GET /api/companies/company/buses/{id}/

Allowed Access: Company Owner only (must own the bus's company)

URL Parameters:

- id (integer, required): Bus ID

Response (200 OK):

json

```
{
  "id": 5,
  "company": 3,
  "name": "Bus 101 - VIP Express",
  "type": "VIP",
  "seat_count": 50,
```

```
"created_at": "2026-01-15T10:30:00Z"
}
```

Error Responses:

- **401 Unauthorized:** Missing or invalid JWT token
 - **403 Forbidden:** User is not a company owner or doesn't own this bus
 - **404 Not Found:** Bus does not exist
-

Endpoint 4: Update Bus (Full Update)

PUT /api/companies/company/buses/{id}/

Allowed Access: Company Owner only (must own the bus's company)

Request Body: (all fields required)

```
json
{
  "name": "Bus 101 - VIP Express Updated",
  "type": "VIP Plus",
  "seat_count": 52
}
```

Response (200 OK): Same as retrieve response with updated data

Error Responses:

- **400 Bad Request:** Validation errors
- **401 Unauthorized:** Missing authentication
- **403 Forbidden:** Not a company owner or doesn't own this bus
- **404 Not Found:** Bus does not exist

Important Notes:

- Updating seat_count may affect existing trips
- Consider implications for trips already scheduled with this bus

Endpoint 5: Partial Update Bus

PATCH /api/companies/company/buses/{id}/

Allowed Access: Company Owner only (must own the bus's company)

Request Body: (partial fields only)

json

```
{  
  "type": "Super VIP",  
  "seat_count": 55  
}
```

Response (200 OK): Same as retrieve response with updated fields

Error Responses: Same as PUT

Endpoint 6: Delete Bus

DELETE /api/companies/company/buses/{id}/

Allowed Access: Company Owner only (must own the bus's company)

Response (204 No Content): (empty body)

Important Notes:

- Buses with existing trips cannot be deleted (foreign key constraint)
- Consider marking as inactive instead of deleting
- Delete will fail if bus is referenced by any trip

Error Responses:

- **401 Unauthorized:** Missing authentication
- **403 Forbidden:** Not a company owner or doesn't own this bus
- **404 Not Found:** Bus does not exist
- **405 Method Not Allowed:** If bus has associated trips

Bus Types Examples

Type	Typical Seat Count	Features
Standard	44 seats	Basic comfort, AC
VIP	30-36 seats	Extra legroom, snacks
Luxury	20-24 seats	Reclining seats, entertainment
Economy	50-60 seats	Budget option

Note: Bus types are not strictly enumerated in the serializer - they can be any string value as defined in the model's choices.

Pagination Information

Default Settings:

- Page size: 20 items per page
- Page parameter: ?page=2
- Custom page size: ?page_size=50

Response Headers:

- count: Total number of buses in company
 - next: URL to next page (or null)
 - previous: URL to previous page (or null)
 - results: Array of bus objects
-

Complete API Endpoints Summary

Method	Endpoint	Authentication	Role	Description
GET	/api/companies/company/buses/	JWT + Owner	Owner	List company buses
POST	/api/companies/company/buses/	JWT + Owner	Owner	Create new bus
GET	/api/companies/company/buses/{id}/	JWT + Owner	Owner	Get single bus
PUT	/api/companies/company/buses/{id}/	JWT + Owner	Owner	Full update bus
PATCH	/api/companies/company/buses/{id}/	JWT + Owner	Owner	Partial update bus
DELETE	/api/companies/company/buses/{id}/	JWT + Owner	Owner	Delete bus

HTTP Status Codes Summary

Status Code	Meaning	When Used
200 OK	Success	GET, PUT, PATCH operations
201 Created	Resource created	Successful bus creation
204 No Content	Success (no body)	Successful deletion

Status Code	Meaning	When Used
400 Bad Request	Client error	Validation failure
401 Unauthorized	Authentication required	Missing/invalid JWT
403 Forbidden	Permission denied	Not a company owner
404 Not Found	Resource not found	Bus doesn't exist

Example API Calls (cURL)

List Company Buses

```
bash
```

```
curl -X GET
```

```
"http://localhost:8000/api/companies/company/buses/?page=1&page_size=20" \
```

```
-H "Authorization: Bearer eyJhbGciOiJIUzI1NiIs..."
```

Create New Bus

```
bash
```

```
curl -X POST http://localhost:8000/api/companies/company/buses/ \
```

```
-H "Authorization: Bearer eyJhbGciOiJIUzI1NiIs..." \
```

```
-H "Content-Type: application/json" \
```

```
-d '{
```

```
  "name": "Bus 202 - City Express",
```

```
  "type": "Standard",
```

```
  "seat_count": 44
```

```
}'
```

Get Single Bus

bash

```
curl -X GET http://localhost:8000/api/companies/company/buses/5/ \
-H "Authorization: Bearer eyJhbGciOiJIUzI1NiIs..."
```

Update Bus (Partial)

bash

```
curl -X PATCH http://localhost:8000/api/companies/company/buses/5/ \
-H "Authorization: Bearer eyJhbGciOiJIUzI1NiIs..." \
-H "Content-Type: application/json" \
-d '{
  "type": "VIP",
  "seat_count": 36
}'
```

Delete Bus

bash

```
curl -X DELETE http://localhost:8000/api/companies/company/buses/5/ \
-H "Authorization: Bearer eyJhbGciOiJIUzI1NiIs..."
```

Important Business Rules

1. Company Isolation:

- Company owners can only see and manage buses belonging to their company
- Automatic filtering via `get_queryset()`
- No cross-company access possible

2. Bus Validation:

- Minimum seat count should be 10 (enforced in Trip creation)
- Bus type can be any string (model likely has choices)

- Bus name should be unique per company

3. Trip Dependencies:

- Buses with existing trips cannot be deleted
- Updating seat_count may affect available seats calculation
- Consider impact on future trips when modifying buses

4. Automatic Company Assignment:

- When creating a bus, company is automatically set from authenticated user
- No need to provide company ID in request

5. Permission Requirements:

- All endpoints require IsAuthenticated + IsCompanyOwner
- Regular users cannot access any bus endpoints
- Even viewing buses is restricted to company owners

Data Model Relationships

text

Company (Owner)

↓

└ Bus

↓

└ Trip

↓

└ Booking

↓

└ Passenger

Key Constraints:

- One company can have multiple buses

- One bus can have multiple trips
 - Bus belongs to exactly one company
 - Bus deletion prevented if trips exist
-

Best Practices for Company Owners

1. Bus Naming Convention:

- Use descriptive names: "Bus 101 - VIP Express"
- Include bus number and type for easy identification

2. Seat Count Management:

- Ensure accurate seat counts for ticket sales
- Consider bus configuration when setting seat_count
- VIP/Luxury buses typically have fewer seats

3. Bus Maintenance:

- Don't delete buses with upcoming trips
- Create new bus entries for different configurations
- Update bus details before scheduling trips

4. Integration with Trips:

- Create buses before scheduling trips
 - Each trip requires a bus assignment
 - Bus availability affects trip scheduling
-

Version Information

- **API Version:** 1.0
- **Last Updated:** May 2026
- **Authentication:** JWT required for all endpoints
- **Role Required:** Company Owner only

- **Pagination:** Enabled (20 items per page default)
-

Complete System API Summary

Now that all apps are documented, here's the complete API structure:

Authentication & Accounts (/api/accounts/)

- Register user/owner
- Login and token management

Trips Management (/api/trips/company/trips/)

- Public trip viewing and search
- Owner-only trip CRUD operations

Bookings (/api/bookings/)

- User booking creation
- Booking listing and cancellation
- Statistics and filtering

Companies/Buses (/api/companies/company/buses/)

- Owner-only bus management
- CRUD operations for company buses